

Sasinathan Rangasamy

Bangalore,Karnataka —  +91-7904404579 —  sasinathan4950@gmail.com
 linkedin.com/in/sasinathan4950

Professional Summary

Embedded Software Engineer with experience in Embedded C, firmware development, STM32, PIC16F877A, ARM Cortex-M, peripheral driver development, FreeRTOS, UART, SPI, I2C, CAN, JTAG debugging, and embedded software testing. Familiar with Git, Linux, Vector CANoe, CANalyzer, and the software development lifecycle. Seeking an opportunity in automotive embedded software and firmware development.

Technical Skills

Programming	Embedded C, C, Python
Microcontrollers	STM32, PIC16F877A, ARM Cortex-M
Protocols	UART, SPI, I2C, CAN
Embedded Systems	Firmware Development, Device Drivers, GPIO, Timers, Interrupts
RTOS	FreeRTOS
Debugging	JTAG, Firmware Debugging
Automotive Tools	Vector CANoe, Vector CANalyzer
Development Tools	MPLAB X IDE, Proteus, Arduino IDE, Git, GitHub, Jira
Operating Systems	Linux, Windows

EXPERIENCE

Data Analyst – Data Annotation Specialist Continental Autonomous Mobility

August 2022 – Present
Bangalore, India

- Delivered high-precision 2D annotations for lanes, vehicles, pedestrians, and road infrastructure, enabling improved ADAS perception model performance.
- Annotated traffic signs, traffic lights, and pavement markings to enhance navigation accuracy and traffic rule recognition systems.
- Performed 3D cuboid annotation on LiDAR point cloud datasets for autonomous driving projects, including ILD and Aurora.
- Utilized Kognic to create accurate 3D bounding boxes and lane annotations for vehicles, pedestrians, cyclists, and road structures.
- Developed Python-based automation scripts for JSON validation, object count verification, data consistency checks, and error reporting, reducing manual quality assurance efforts.
- Conducted tool validation, root cause analysis, and corrective labeling to maintain high-quality AI/ML training datasets and ensure compliance with annotation guidelines.
- Analyzed large-scale annotation datasets to identify labeling inconsistencies and implemented process improvements for enhanced data quality.
- Contributed to the creation of robust training datasets supporting object detection, tracking, lane detection, and autonomous vehicle perception models.

Projects

Smart Irrigation System using PIC16F877A

Academic Project

Designed and developed a smart irrigation system using the **PIC16F877A** microcontroller and **Embedded C** to automate water pump control based on soil moisture levels. Interfaced soil moisture sensors, relay modules and a 16×2 LCD using GPIO, implemented UART communication for monitoring and debugging, and validated the firmware using **MPLAB X IDE**, **XC8 Compiler** and **Proteus**.

Education

B.E. Electronics and Communication Engineering
Sri Sai Ranganathan Engineering College

2022
CGPA: 8.26/10

Certification

Advanced Embedded Systems Certification
Bytes In Bits Technologies

2026
Bangalore